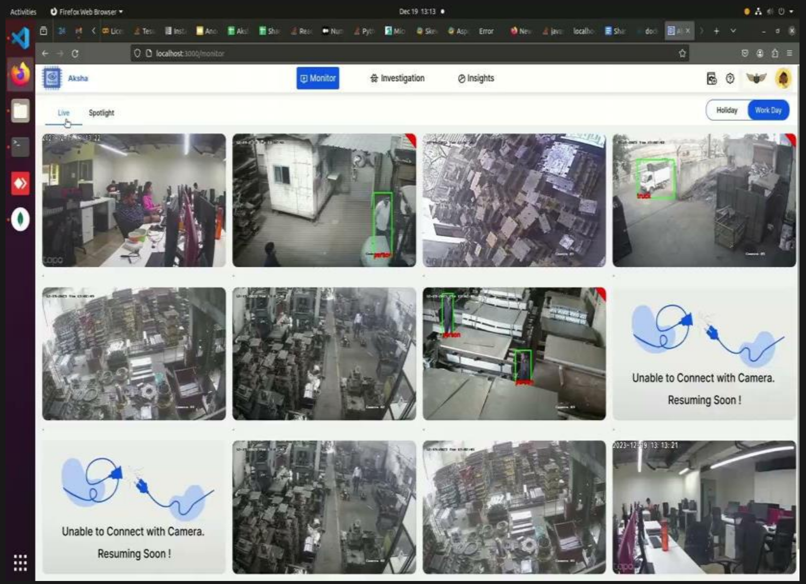
Introduction to Aksha

Advanced AI Solutions for Manufacturing Excellence

Overview of Aksha

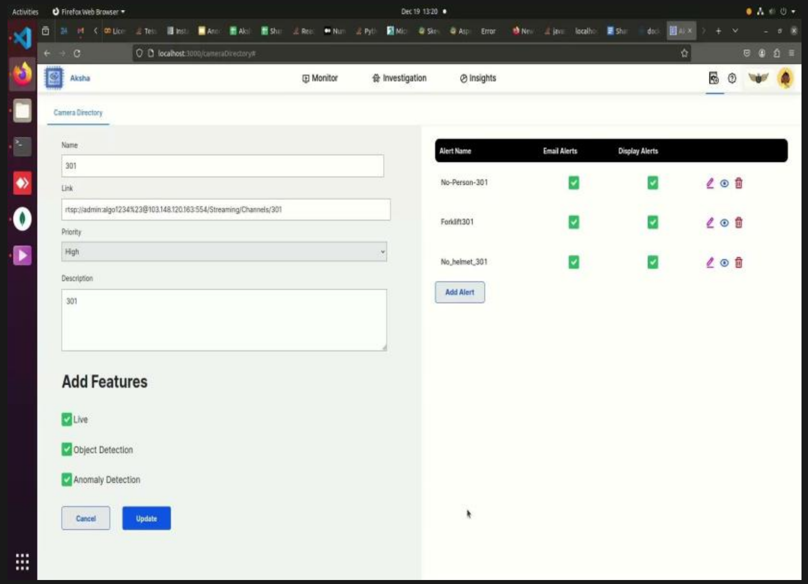
- Aksha is an intelligent surveillance system designed for automated monitoring with minimal human intervention.
- Utilizes advanced technologies such as Computer Vision, Artificial Intelligence, and Generative AI for real-time analysis.
- Key functionalities include object detection, anomaly identification, and automated response mechanisms.

Architectural Approaches



- Multiple deployment models: Aksha on Edge (On-Premise), Aksha UI & Backend on Edge, and hybrid models combining Edge & Cloud.
- Supports seamless integration with existing CCTV systems through standard RTSP links.
- Decoupled architecture enhances maintainability and fault tolerance, allowing for independent service development.

Key Features of Aksha



- Live recording and processing capabilities for immediate data access.
- Advanced object detection including various categories such as vehicles, personnel, and safety gear compliance.
- Customizable alerts enable users to set specific notifications based on detected anomalies.

Collaboration and Customization

- Partnership with NDIL to enhance Aksha for worker safety, integrating Edge + Cloud solutions.
- Phased development approach, with significant enhancements in Gen AI capabilities for improved anomaly detection.
- Successful proof of concept executed on Raspberry Pi 4B, showcasing versatility and cost-effectiveness.

Customer Testimonials

- Clients report significant reductions in human monitoring efforts and improved detection of unforeseen events.
- Positive feedback highlights Aksha's ability to transform traditional CCTV usage into proactive safety management.
- Ongoing interest in future updates and innovations, indicating strong customer engagement and satisfaction.

Scalability and Performance

- Effortless scalability achieved through microservices architecture and Kubernetes orchestration.
- Real-time data processing capabilities ensure timely insights and actions for enhanced operational efficiency.
- High availability maintained through distributed messaging systems like Kafka, ensuring reliable performance.